

Daniel Ye

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EDUCATION

Columbia University - Fu Foundation School of Engineering

New York City, NY

B.S. in Electrical Engineering, GPA: 3.66/4.0

Aug 2024 – May 2028

- **Related Coursework:** Digital VLSI, Analog Logic Design, Electronic Circuit, Electromagnetic, Communication System, FUNDIES, Solid State Devices, Circuit Analysis, Signals & Systems, Discrete Math, Data Structures
- **Teaching Assistant:** Art of Engineering Common Project (Fall 2025, Spring 2026, Fall 2027)

EXPERIENCES

The Columbia Laboratory for Unconventional Electronics - Researcher

Sep 2025 – Present

- Advanced pre-contact awareness in robotic hands by cadding a motor-controlled parallel gripper with capacitive proximity sensors for proximity-guided grasping. Designed a custom PCB, featuring a Cypress CapSense microcontroller, to integrate power regulation, sensor routing, and external filtering capacitors, minimizing motor EMI and baseline capacitance drift for stable sensor performance.
- Programmed firmware in ModusToolbox to detect metallic and ionic targets, streaming real-time proximity data through UART communication. Implemented a Python control pipeline that parses the UART data, using PySerial, to control two Dynamixel motors with an exponential decay velocity profile.

Amazon - Software Development Engineer Intern

May 2026 – Aug 2026

- Helped decision-making on delaying package arrival dates by building a two-stage LightGBM model that estimates customer order loss from weather events across 16,000 U.S. zip codes. Consolidated necessary datasets, optimized features, trained/test model, and visualized predictions on an interactive choropleth map.
- Productionized an automated ML pipeline using AWS CDK, orchestrating scheduled ingestion and inference across Redshift, Lambda, and EventBridge with containerized execution on ECS Fargate.

Columbia Robotics Club - CCBR Head of Electrical Engineering, Business Chair

Sep 2025 - Present

- Improved power delivery reliability for a robotic quadruped by designing its power distribution schematic and manufacturing/soldering the custom PCB.
- Enhanced onboard perception for quadruped locomotion by configuring the NVIDIA Jetson Nano and designing a multi-sensor PCB integrating an IMU, magnetic encoders, and other sensors.

Columbia Space Initiative - Balloons Electrical Engineer

Aug 2024 – May 2025

- Enabled long-duration high-altitude flight tracking by designing a custom PCB in KiCad, integrating environmental sensors, a GPS tracker, and a Pico-series microcontroller; manufactured boards through JLCPCB and assembled picoballoon payloads.
- Supported successful deployment and real-time tracking of a high-altitude balloon in upstate New York through collaborative construction, sensor calibration, and communications testing.

Amazon - Software Development Engineer Intern

May 2025 – Aug 2025

- Improved user interaction and data visualization for an optimization tool generating 4,000+ daily executions by enhancing a web-based history dashboard and developing an execution analysis page using React and Typescript.
- Enabled efficient data retrieval and management by building REST APIs and backend services in Python with AWS (S3, API Gateway, Lambda, CloudWatch) and configuring an SQL RDS database.

PROJECTS

Smart Trash

Nov 2025

- In a team of four, developed a computer-vision recycling system on Raspberry Pi that automatically identifies recyclable vs. landfill waste and mechanically sorts items; fabricated a full prototype using laser-cut plywood and launched a Vultr-deployed real-time monitoring website; awarded Best Use of Vultr and Best Pitch.

Rock Paper Scissors Game

Dec 2024

- Designed and built a fully functional two-player Rock Paper Scissors game in hardware using 74HC00N NAND gate ICs, implementing move selection logic, mutual exclusion, and win/draw detection entirely in combinational logic on a breadboard.
- Minimized the win-condition logic circuit using Karnaugh maps, then simulated the full system in Logically before physical assembly.

VHDL Automatic Smart Player

Apr 2026

- Implemented automatic playback of a song on a Basys 3 FPGA in VHDL; designed a tempo generation process using a 1 MHz clock divider, a 29-step FSM-based note sequencer, and note-to-frequency clock dividers; used Vivado for waveform simulation and debugging; added onboard LED indicators and a 7-segment display for real-time playback feedback.

SKILLS

Software Languages: Python, Java, Typescript with React, C++, Assembly, SystemVerilog, VHDL

Technology: Linux, Git, KiCad, Logically, LTspice, Vivado, Fusion Autodesk, MATLAB, Oscilloscope, Multimeter, AWS